Weekly training log, week 10

Week 10

Module	EEEM068, Applied Machine Learning
Project	Industrial waste classification on WaRP
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Focus	YOLO26 Small and Medium on WaRP-D

1. Experimental settings

Dataset	WaRP-D, 2452 train, 522 val, 28 classes
GPU	NVIDIA RTX 4000 Ada, 20 GB
Framework	Ultralytics 8.4.41
Metric	COCO mAP at IoU 0.5 and mAP from 0.5 to 0.95
Goal	One stage real time detector on WaRP-D.

2. Hyperparameters

Trial 1, YOLO26 Small (12.5 M params)

Weights	yolo26s.pt
Image size	640 x 640
Batch size	16
Epochs	100 (patience 20)
Optimizer	SGD, lr0 0.01, lrf 0.01
Momentum	0.937
Weight decay	5e-4
Warmup epochs	3
Mosaic / MixUp	1.0 / 0.5 (close_mosaic last 10 epochs)
Erasing	0.4
HSV (h, s, v)	0.015 / 0.7 / 0.4
Seed	42

Trial 2, YOLO26 Medium (20 M params)

Weights	yolo26m.pt
Change	Only the backbone size.

3. Results

Trial 1, YOLO26 Small

Epoch	box	cls	dfl	P	R	mAP50	mAP50-95
10	1.013	1.151	1.004	0.671	0.539	0.535	0.388
30	0.737	0.605	0.822	0.785	0.652	0.604	0.452
50	0.630	0.421	0.747	0.808	0.661	0.601	0.458
80	0.549	0.298	0.697	0.811	0.660	0.602	0.458
100	0.524	0.268	0.683	0.691	0.614	0.604	0.471

Best mAP@50: 60.4%. mAP 50 to 95: 47.1%.

Trial 2, YOLO26 Medium

Epoch	box	cls	dfl	P	R	mAP50	mAP50-95
10	0.978	1.088	0.986	0.661	0.557	0.561	0.412
30	0.711	0.572	0.808	0.781	0.665	0.625	0.471

Epoch	box	cls	dfl	P	R	mAP50	mAP50-95
50	0.604	0.395	0.734	0.808	0.671	0.638	0.494
75	0.541	0.310	0.708	0.812	0.660	0.638	0.494
100	0.498	0.252	0.681	0.691	0.614	0.638	0.493

Best mAP@50: 63.8%. mAP 50 to 95: 49.4%.

4. Observations

Going from small to medium gave another 3.4 mAP@50. Worth the extra params.

The small precision drop at the end is just close_mosaic kicking in for the last 10 epochs, no real impact on mAP.

Hardest classes are detergent-transparent (AP 0.32) and juice-cardboard (AP 0.41), mostly missed entirely. Full bottles always do better than crushed ones.

Submitting YOLO26 Medium with 63.8% mAP@50 as our detection result.